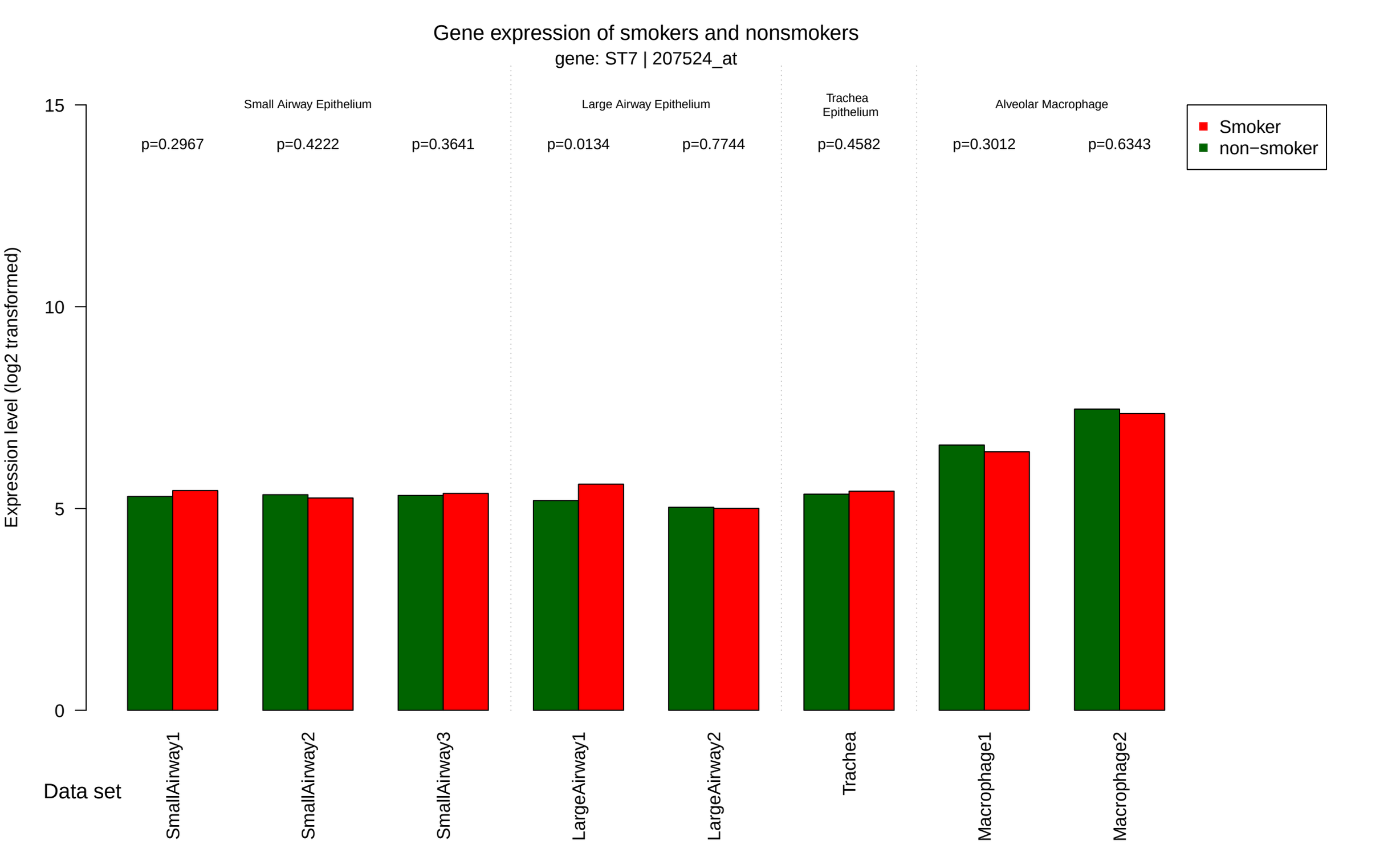
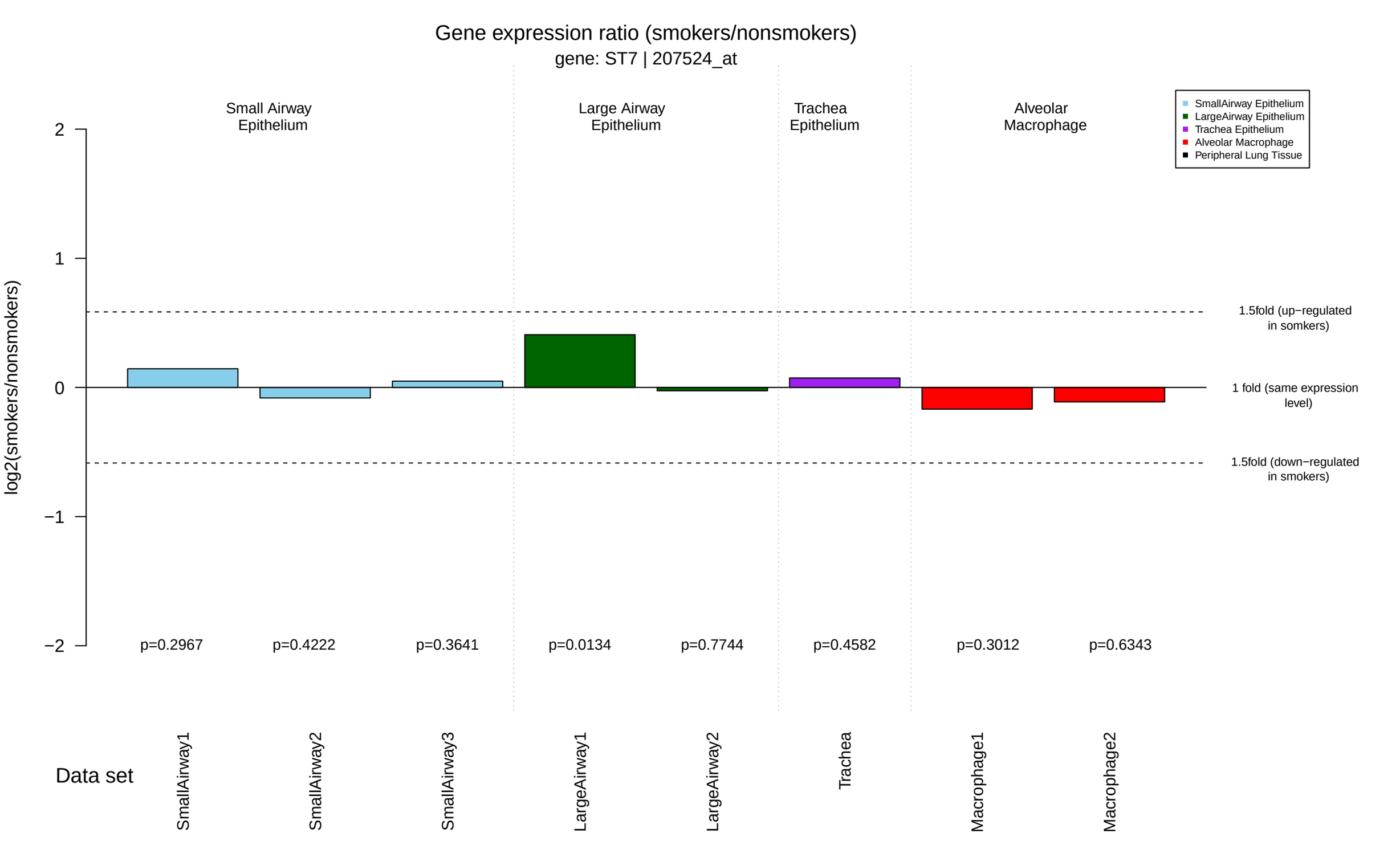
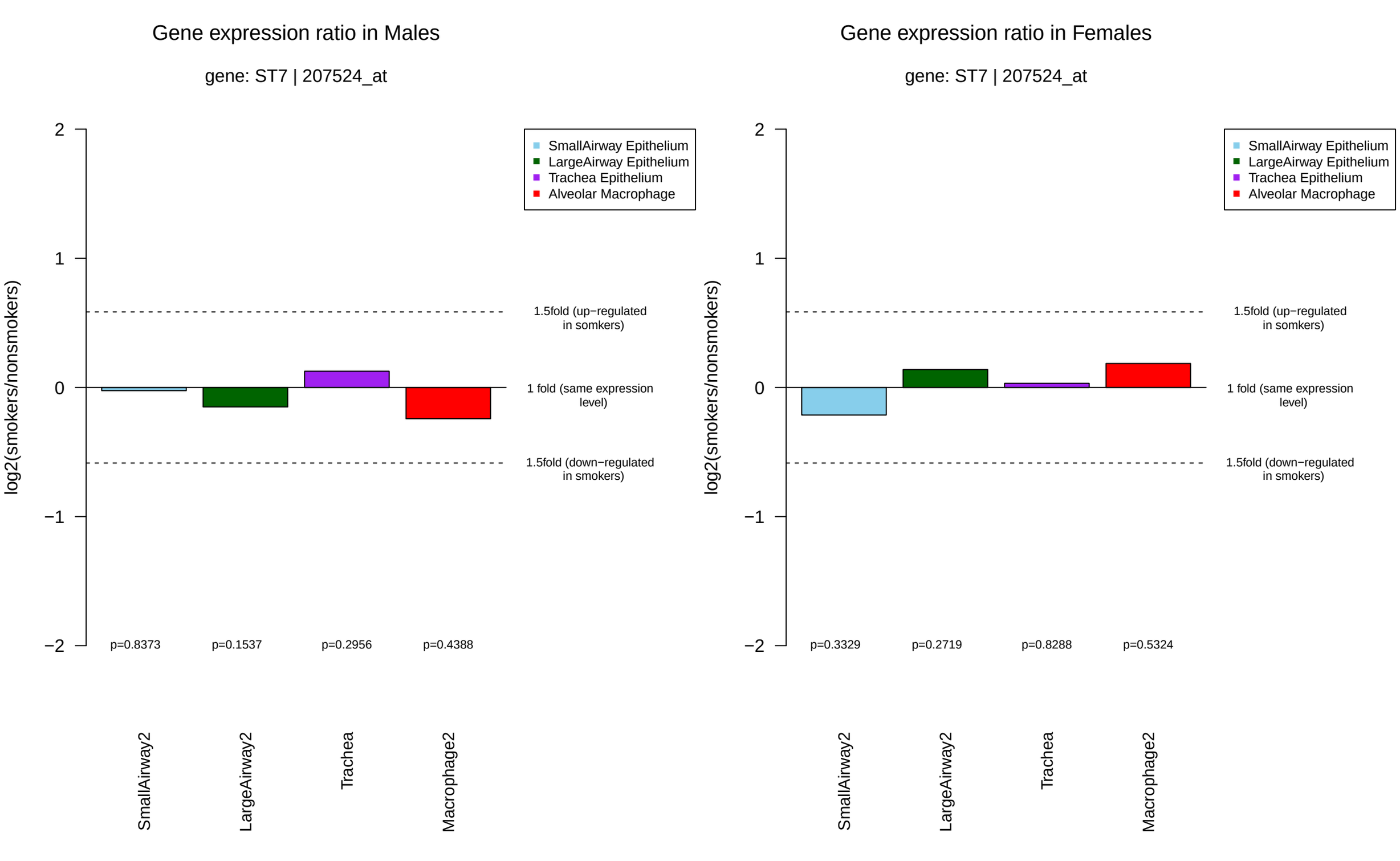
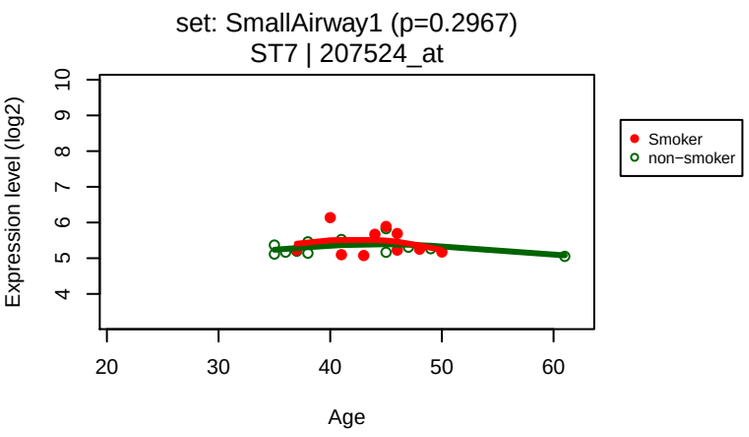
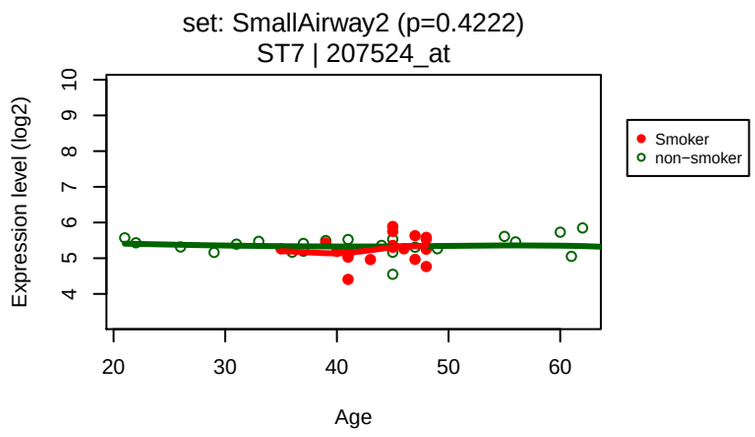
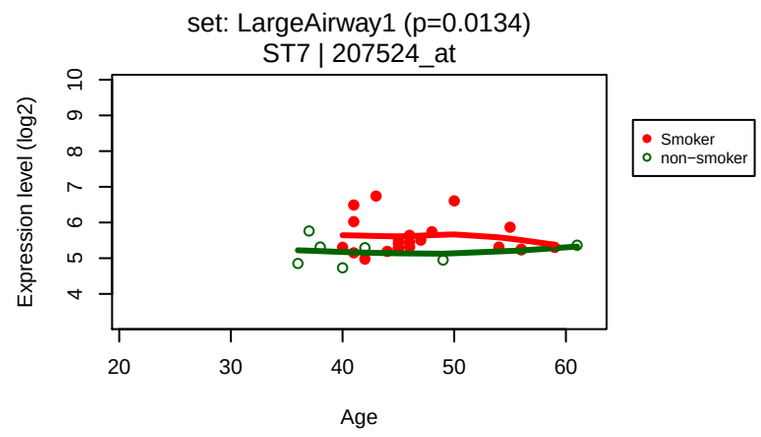

Gene expression of smokers and nonsmokers
gene: ST7 | 207524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2967
p=0.4222
p=0.3641
p=0.0134
p=0.7744
p=0.4582
p=0.3012
p=0.6343
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

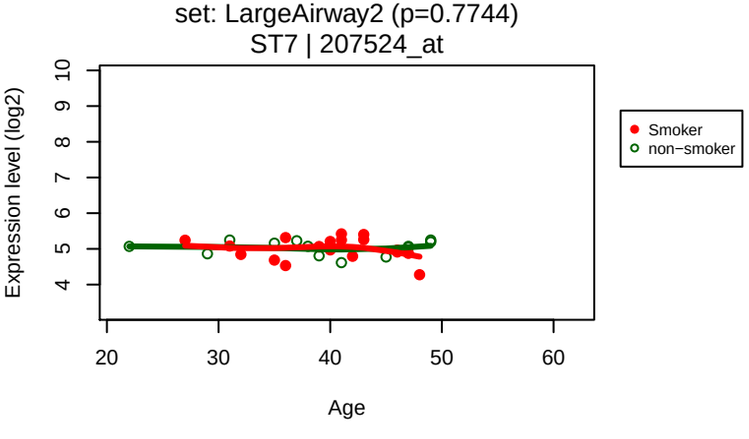

Gene expression ratio (smokers/nonsmokers)
gene: ST7 | 207524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2967
p=0.4222
p=0.3641
p=0.0134
p=0.7744
p=0.4582
p=0.3012
p=0.6343
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

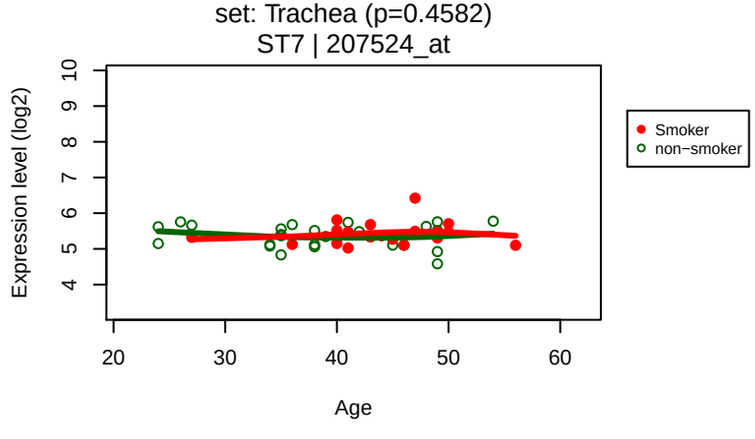

Gene expression ratio in Males
gene: ST7 | 207524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8373
p=0.1537
p=0.2956
p=0.4388
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST7 | 207524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3329
p=0.2719
p=0.8288
p=0.5324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

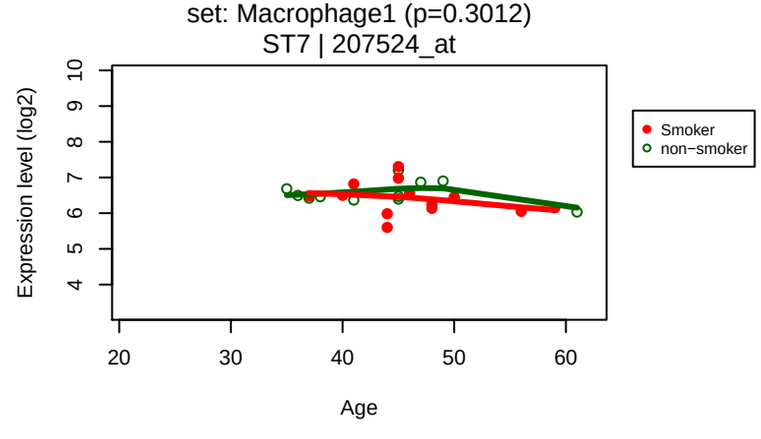

set: SmallAirway1 (p=0.2967)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker

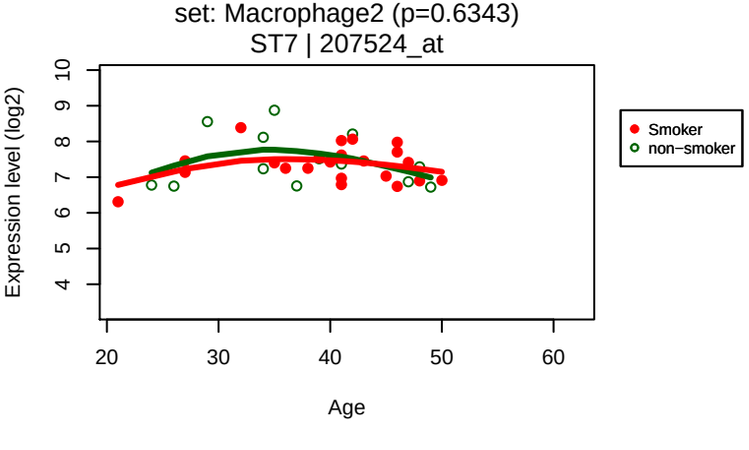

set: SmallAirway2 (p=0.4222)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker

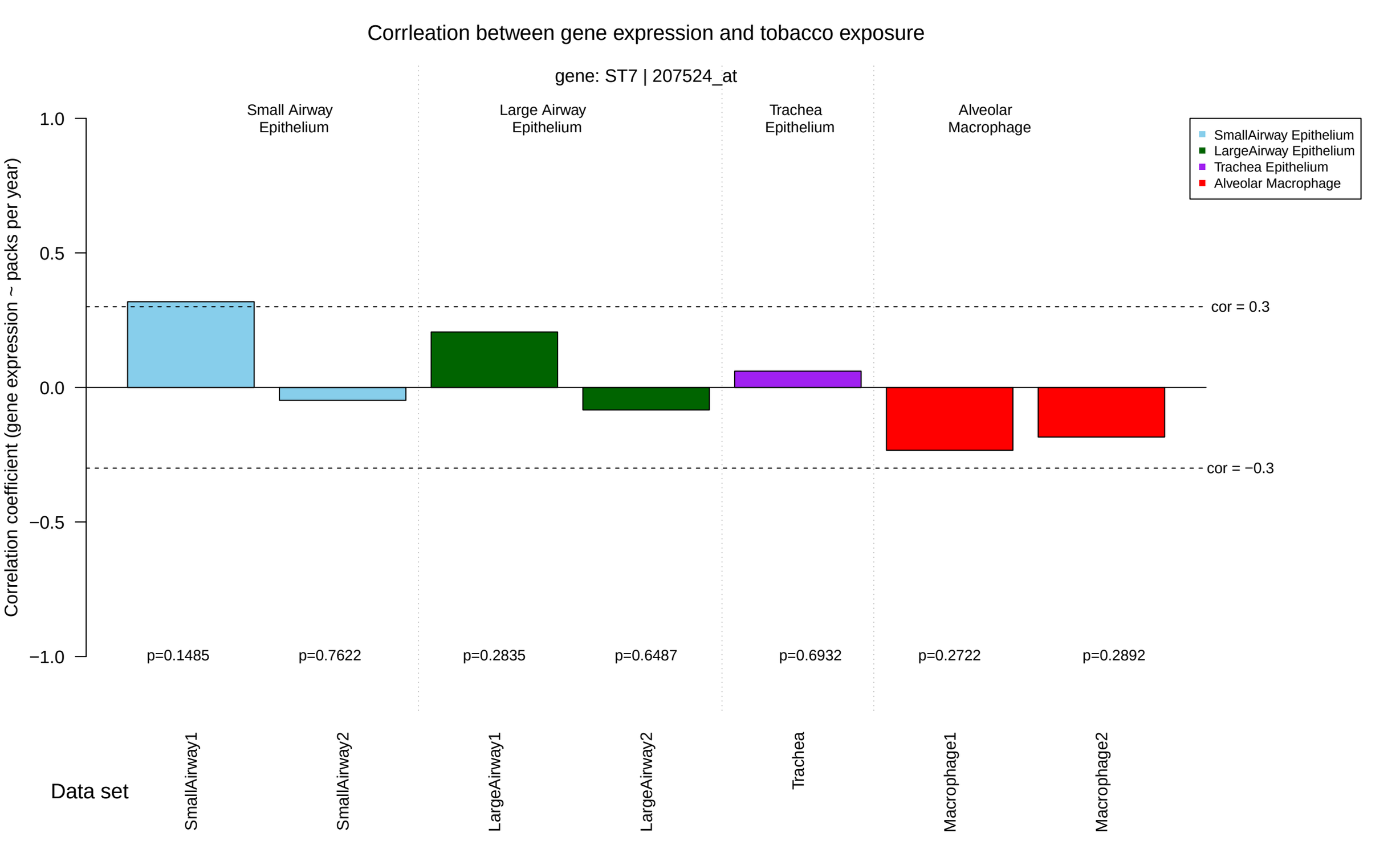

set: LargeAirway1 (p=0.0134)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7744)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4582)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3012)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6343)
ST7 | 207524_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST7 | 207524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1485
p=0.7622
p=0.2835
p=0.6487
p=0.6932
p=0.2722
p=0.2892
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2